Saviours AI

ICSE CLASS 10 · 2027 SYLLABUS

# Physics Cheat Sheet

Every formula, law and unit from ICSE Class 10 Physics.
8 pages. Nothing missing.

Moments | Work & Energy | Machines | Light | Sound | Electricity | Magnetism | Heat | Radioactivity

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## 01 Units, constants & conversions

### Common derived units

| Quantity | SI unit | In base units |
|---|---|---|
| Force | newton (N) | kg m $s^{-2}$ |
| Work / Energy | joule (J) | N m |
| Power | watt (W) | J $s^{-1}$ |
| Pressure | pascal (Pa) | N $m^{-2}$ |
| Moment (torque) | N m | N × m |
| Frequency | hertz (Hz) | $s^{-1}$ |
| Charge | coulomb (C) | A s |
| Potential diff. | volt (V) | J $C^{-1}$ |
| Resistance | ohm (Ω) | V $A^{-1}$ |
| Heat capacity | J $K^{-1}$ | |
| Specific heat | J $kg^{-1}$ $K^{-1}$ | |
| Sp. latent heat | J $kg^{-1}$ | |
| Power of lens | dioptre (D) | $m^{-1}$ |

### Prefixes

| Prefix | Symbol | Factor |
|---|---|---|
| giga | G | $10^9$ |
| mega | M | $10^6$ |
| kilo | k | $10^3$ |
| centi | c | $10^{-2}$ |
| milli | m | $10^{-3}$ |
| micro | μ | $10^{-6}$ |
| nano | n | $10^{-9}$ |

### Conversions you need

| | |
|---|---|
| 1 cal | 4.186 J |
| 1 kW h | $3.6 \times 10^6$ J |
| 1 hp | 746 W |
| 1 eV | $1.6 \times 10^{-19}$ J |
| 1 erg | $10^{-7}$ J |
| 1 dyne | $10^{-5}$ N |
| 1 kgf | 9.8 N |
| T (K) | t °C + 273 |
| 1 g $cm^{-3}$ | 1000 kg $m^{-3}$ |
| 1 atm | $1.013 \times 10^5$ Pa |

## 02 Class IX foundations you still use

### Motion in a straight line

| | |
|---|---|
| Speed | distance / time |
| Velocity | displacement / time |
| Acceleration | $a = (v - u)/t$ |

$$v = u + at$$

$$s = ut + \tfrac{1}{2}at^2$$

$$s = \tfrac{1}{2}(u + v)t$$

$$v^2 = u^2 + 2as$$

- **Graphs:** slope of s–t = velocity; slope of v–t = acceleration; area under v–t = displacement.

### Newton's laws & gravitation

| | |
|---|---|
| Momentum | $p = mv$ |
| 2nd law | $F = ma = \Delta p/\Delta t$ |
| Weight | $W = mg$ |
| Gravitation | $F = G\, m_1 m_2 / r^2$ |
| G | $6.67 \times 10^{-11}$ N $m^2$ $kg^{-2}$ |

- **1st law:** a body stays at rest or in uniform motion unless a net force acts (inertia).
- **3rd law:** action and reaction are equal, opposite and act on **different** bodies.

### Fluids: pressure, buoyancy

| | |
|---|---|
| Pressure | $P = F / A$ |
| Liquid column | $P = h\varrho g$ |
| Upthrust | $F_B = V\, \varrho_{fluid}\, g$ |
| Relative density | $\varrho_{sub} / \varrho_{water}$ |

- **Pascal's law:** pressure applied on an enclosed liquid is transmitted equally in all directions.
- **Archimedes:** upthrust = weight of fluid displaced.
- **Floats** if $\rho_{body} < \rho_{liquid}$. **Sinks** if greater. **Floats fully, neutral** if equal.

| | |
|---|---|
| Floatation | $\varrho_1/\varrho_2 = V_2/V_1$ |

### Plane & spherical mirrors

| | |
|---|---|
| Mirror formula | $1/v + 1/u = 1/f$ |
| Focal length | $f = R / 2$ |
| Magnification | $m = -v/u = h'/h$ |

- **Sign convention:** distances from the pole; along the incident ray +, opposite −. Concave: f is −. Convex: f is +.
- **Plane mirror:** image is virtual, erect, same size, laterally inverted, as far behind as the object is in front.

| | |
|---|---|
| Two mirrors at θ | $n = 360°/\theta - 1$ |

### Sound & simple pendulum

$$v = f\lambda$$

| | |
|---|---|
| Frequency | $f = 1 / T$ |
| Pendulum | $T = 2\pi\sqrt{l/g}$ |
| Echo | $2d = v\,t$ |

- **l vs $T^2$** graph is a straight line through the origin; slope = $4\pi^2/g$.
- **Sound** needs a medium. It cannot travel in vacuum. Light can.

## 03 Force, moments & circular motion

### Moment of a force (torque)

$$\text{Moment} = F \times d$$

| | |
|---|---|
| SI unit | newton metre (N m) |
| CGS unit | dyne cm |

- **d** is the perpendicular distance from the pivot to the line of action of the force.
- **Clockwise** moment is negative, **anticlockwise** is positive (by convention).
- Examples: opening a door, steering wheel, bicycle pedal, spanner, see-saw.

### Equilibrium of a body

- **Translational:** net force = 0 ($\Sigma F = 0$).
- **Rotational:** sum of clockwise moments = sum of anticlockwise moments.

**Principle of moments: clockwise = anticlockwise**

| | |
|---|---|
| Balance at pivot | $F_1 d_1 = F_2 d_2$ |

- **Verification:** metre rule hung from two spring balances with slotted weights; the readings obey $\Sigma M = 0$ about any point.

### Uniform circular motion

- Constant **speed**, changing **velocity**, so there is acceleration towards the centre.

| | |
|---|---|
| Direction of velocity | tangent to the circle |

- **Centripetal force:** real force, acts towards the centre. Keeps the body on the circle. Examples: string tension, gravity on a satellite, friction on a turning car.
- **Centrifugal force:** apparent (pseudo) force felt in a rotating frame, acts away from the centre.

| | Centripetal | Centrifugal |
|---|---|---|
| **Direction** | Towards centre | Away from centre |
| **Nature** | Real force | Pseudo force |
| **Does work?** | No (⊥ to motion) | Not a real force |

## 04 Work, energy & power

### Work

$$W = F\,s\cos\theta$$

| | |
|---|---|
| θ = 0° | W = F s (maximum) |
| θ = 90° | W = 0 |
| θ = 180° | W = −F s |
| Against gravity | W = m g h |
| SI unit | joule (J) = N m |
| CGS unit | erg (1 J = $10^7$ erg) |

### Energy

- Energy is the **capacity to do work**. Same unit as work: joule.

$$KE = \tfrac{1}{2}\,m\,v^2$$

$$PE = m\,g\,h$$

- **KE derivation:** $v^2 = u^2 + 2as$ with u = 0, so $s = v^2/2a$. $W = F\,s = ma \times v^2/2a = \tfrac{1}{2}mv^2$.

| | |
|---|---|
| Work–energy | $W = \Delta KE$ |
| Momentum link | $KE = p^2 / 2m$ |

### Power

$$P = W / t = F\,v$$

| | |
|---|---|
| SI unit | watt (W) = J $s^{-1}$ |
| 1 kW | $10^3$ W |
| 1 MW | $10^6$ W |
| 1 GW | $10^9$ W |
| 1 hp | 746 W |
| Commercial unit | 1 kW h = 3.6 × $10^6$ J |

### Conservation of energy

$$U + K = \text{constant}$$

- **Free fall from height h:** at top U = mgh, K = 0. At any point U + K = mgh. At the bottom U = 0 and K = ½mv$^2$ = mgh.

Speed on landing — $v = \sqrt{2gh}$

- **Simple pendulum:** at the extremes, all PE; at the mean position, all KE. Total stays constant (ignoring air).

## 05 Simple machines

### Key terms & formulas

| | |
|---|---|
| Mechanical advantage | $MA = L / E$ |
| Velocity ratio | $VR = d_E / d_L$ |
| Efficiency | $\eta = W_{out}/W_{in} \times 100\%$ |
| η in MA, VR | $\eta = MA / VR \times 100\%$ |
| Ideal machine | $\eta = 100\%$, $MA = VR$ |

- **L** = load, **E** = effort. $d_E$ = distance moved by effort, $d_L$ = by load.
- **Real machine:** η < 100% and MA < VR, because of friction and weight of moving parts.

### Principle of a lever

$$\text{Effort} \times \text{effort arm} = \text{Load} \times \text{load arm}$$

Lever MA — $MA = L/E = d_E/d_L$ (ideal)

VR of lever — VR = effort arm / load arm

- Effort arm longer than load arm: **force multiplier** (MA > 1).

### Three classes of lever

| Class | Fulcrum position | MA | Examples |
|---|---|---|---|
| I | Between load and effort | >1, =1 or <1 | Scissors, see-saw, crowbar, pliers |
| II | Load in the middle | Always > 1 | Wheelbarrow, nutcracker, bottle opener |
| III | Effort in the middle | Always < 1 | Forearm, tongs, fishing rod |

### Pulley systems

| Pulley | VR | MA (ideal) | Effort direction |
|---|---|---|---|
| Single fixed | 1 | 1 | Changes direction |
| Single movable | 2 | 2 | Upward, halves the effort |
| Block & tackle (n strands) | n | n | Depends on the set-up |

- **Single fixed:** only changes the direction of effort. **Movable:** supports load with two strands. **Block and tackle:** VR = number of rope segments supporting the load.

## 06 Light: refraction, lenses

### Laws of refraction

- **1.** Incident ray, refracted ray and normal lie in the **same plane**.
- **2.** Snell's law: sin i / sin r = constant (the refractive index).

$$\mu = \sin i / \sin r = c / v = \lambda_{air} / \lambda_{medium}$$

Real / apparent depth — $\mu$ = real / apparent

- **Frequency never changes** on refraction. Speed and wavelength change.

### Refractive indices

| Medium | μ |
|---|---|
| Air / vacuum | 1.00 |
| Water | 1.33 |
| Glass | 1.5 |
| Diamond | 2.42 |

- **Lateral shift:** sideways displacement of the ray on passing through a glass slab.
- **Emergent ray is parallel to the incident ray** through a slab (emergence angle = incidence angle).

### Total internal reflection (TIR)

Critical angle — $\mu = 1 / \sin C$

- **Conditions:** (1) ray goes from **denser to rarer**; (2) angle of incidence **> C**.

C for glass — ≈ 42°

C for water — ≈ 49°

C for diamond — ≈ 24°

- **Uses:** right-angle prism (bends light 90° or 180°), optical fibres, periscope, binoculars, mirage, sparkle of diamond.

### Lens formula

$$1/v - 1/u = 1/f$$

Magnification — $m = v / u = h' / h$

Power — P = 1 / f (f in metres)

Unit of power — dioptre (D)

Combination — $P = P_1 + P_2$

- **Sign convention:** distances from optical centre; along the incident ray +. Convex lens: f is +, P is +. Concave: f is −, P is −.

### Convex lens: image table

| Object at | Image | Nature |
|---|---|---|
| ∞ | At F | Real, inverted, tiny |
| Beyond 2F | Between F and 2F | Real, inv., smaller |
| At 2F | At 2F | Real, inv., same size |
| Between F, 2F | Beyond 2F | Real, inv., larger |
| At F | ∞ | Real, inv., huge |
| Within F | Same side | Virtual, erect, larger |

- **Concave lens:** always virtual, erect and diminished, between O and F.

### Dispersion & spectrum

- **VIBGYOR:** violet bends most (shortest λ), red bends least (longest λ).
- **Why the sky is blue:** short blue λ scatters most. **Sunset red:** blue scatters away, red reaches us.
- **EM spectrum (increasing λ):** γ-rays → X-rays → UV → visible → IR → microwaves → radio waves.
- **All EM waves:** travel at c in vacuum, transverse, need no medium, carry energy.

## 07 Sound waves

### Wave basics

$$v = f \lambda$$

| | |
|---|---|
| Time period | $T = 1 / f$ |
| Speed in air | ~330–340 m $s^{-1}$ |
| Speed order | solids > liquids > gases |

- **Speed increases** with temperature and humidity; independent of pressure (same temperature).
- **Thunder vs lightning:** light ($3 \times 10^8$ m/s) is far faster than sound, so you see lightning first.

### Echo

$$2d = v\,t \rightarrow d = v\,t / 2$$

- **Minimum distance** for a clear echo ≈ **17 m** (persistence of hearing 0.1 s, v = 340 m/s).
- **Uses:** SONAR, bats and dolphins, depth of the sea, medical ultrasound, fishermen.
- **SONAR:** Sound Navigation And Ranging. Ultrasound is sent, reflected and timed.

### Frequency ranges

| Type | Range | Example |
|---|---|---|
| **Infrasonic** | < 20 Hz | Earthquakes, whales |
| **Audible (sonic)** | 20 Hz – 20 kHz | Human hearing |
| **Ultrasonic** | > 20 kHz | Bats, SONAR, scans |

- **Supersonic** means faster than sound (jets). It is **not** the same as ultrasonic.

### Types of vibration

| Type | Meaning | Example |
|---|---|---|
| **Natural / free** | Vibrates at its own frequency, no outside force | Pendulum, tuning fork |
| **Damped** | Amplitude falls due to friction | Spring in air |
| **Forced** | Driven by an external periodic force | Sitar body, tuning fork on table |
| **Resonance** | f(applied) = natural f; amplitude becomes maximum | Glass shatters, sonometer, swing |

### Characteristics of sound

| Property | Depends on | Unit / note |
|---|---|---|
| **Loudness** | Amplitude (and intensity) | decibel (dB) |
| **Pitch** | Frequency | High f = shrill |
| **Quality (timbre)** | Waveform | Tells instruments apart |

| | |
|---|---|
| Intensity | power / area (W $m^{-2}$) |

- **Noise pollution:** above ~80 dB is harmful. Reduce with trees, soundproofing, silence zones.

## 08 Current electricity

### Core relations

| | |
|---|---|
| Current | $I = Q / t$ |
| Potential diff. | $V = W / Q$ |

$$V = I\,R$$

| | |
|---|---|
| Resistance | $R = V / I$ |

- **Ohm's law:** at constant temperature, $V \propto I$.
- **V–I graph** (V on the y-axis, I on the x-axis): straight line through the origin; **slope = R**.
- **Non-ohmic:** filament lamp, diode, thermistor (V–I curve is not straight).

### Resistance depends on

$$R = \varrho\,l / A$$

- **Length** ↑ R ↑. **Area** ↓ R ↑. **Temperature** ↑ R ↑ (metals). **Material** (resistivity ρ).

| | |
|---|---|
| Unit of ρ | ohm metre (Ω m) |

- **Superconductor:** zero resistance below a critical temperature.

## Series & parallel

| | Series | Parallel |
|---|---|---|
| **Current** | Same everywhere | Splits: $I = I_1 + I_2$ |
| **Voltage** | Splits: $V = V_1 + V_2$ | Same across each |
| **Total R** | $R = R_1 + R_2 + \ldots$ | $1/R = 1/R_1 + 1/R_2 + \ldots$ |

Two in parallel — $R = R_1R_2 / (R_1 + R_2)$

n equal in parallel — $R / n$

- **Parallel total R is always smaller** than the smallest resistor.

## EMF & internal resistance

EMF — $E = V + I r$

With load R — $E = I (R + r)$

Internal resistance — $r = (E - V) / I$

- **EMF** = work done per unit charge by the cell in the **whole** circuit (open circuit). **Terminal pd** is less than EMF when current flows.

## Electrical energy & power

$$W = V I t = I^2 R t = (V^2/R) t$$

$$P = V I = I^2 R = V^2 / R$$

Energy — $E = P \times t$

Commercial unit — $1\ kW\ h = 1000\ W \times 1\ h$

Cost — units × rate per unit

Units (kW h) — $W \times hours / 1000$

# 09 Household circuits, magnetism & EMI

## Household wiring

| Wire | Old colour | New colour | Pin |
|---|---|---|---|
| **Live (phase)** | Red | Brown | Right (viewed front) |
| **Neutral** | Black | Blue | Left |
| **Earth** | Green | Green/Yellow | Top, thickest and longest |

- **Fuse and switch go in the live wire.** It breaks the circuit when current exceeds its rating.

Fuse rating — $I = P / V$ (pick the next higher)

- **Earthing:** sends leakage current into the ground, protects from shock. Needed for metal bodies of heavy appliances.
- **MCB** replaces a fuse: trips and can be reset.

## Magnetic effect of current

- **Oersted:** a current-carrying wire deflects a compass needle.
- **Right-hand thumb rule:** thumb along current, curled fingers show the field direction. For a coil, curl fingers along the current and the thumb points to the north pole.

Field strength ↑ — more turns, more current, soft-iron core

- **Electromagnet** vs permanent: can be switched off, strength is adjustable, polarity reversible.

## Fleming's rules

| | Left-hand rule | Right-hand rule |
|---|---|---|
| **Used in** | DC electric motor | AC generator |
| **Thumb** | Force (motion) | Motion |
| **Forefinger** | Field N → S | Field N → S |
| **Middle** | Current | Induced current |
| **Energy** | Electrical → Mechanical | Mechanical → Electrical |

- **Motor parts:** coil, magnet, split-ring commutator, brushes. **Generator:** coil, magnet, slip rings, brushes.

## Electromagnetic induction

- **Faraday:** changing magnetic flux through a coil induces an emf.
- Induced emf is larger with: faster motion, stronger magnet, more turns.

AC frequency (India) — 50 Hz

- **AC over DC:** can be stepped up/down by a transformer, cheap long-distance transmission, less loss.

## Transformer

$$V_s/V_p = N_s/N_p = I_p/I_s$$

Ideal — $V_pI_p = V_sI_s$

Efficiency — $\eta = P_{out}/P_{in} \times 100\%$

| | Step-up | Step-down |
|---|---|---|
| **Turns** | $N_s > N_p$ | $N_s < N_p$ |
| **Voltage** | Increases | Decreases |
| **Current** | Decreases | Increases |
| **Secondary wire** | Thinner | Thicker |

- **Works only on AC.** Core is laminated soft iron to reduce eddy currents.

## 10 Heat & modern physics

### Calorimetry

$$Q = m\,c\,\Delta T$$

| | |
|---|---|
| Heat capacity | $C' = Q / \Delta T = m\,c$ |
| Specific heat | $c = Q / (m\,\Delta T)$ |
| 1 cal | 4.186 J |

- **Method of mixtures:**

**Heat lost by hot = Heat gained by cold**

$m_1c_1(T_1 - T) = m_2c_2(T - T_2)$

- **High c of water:** coolant in engines, sea breeze, hot-water bottles, moderates climate.

### Latent heat (fusion)

$$Q = m\,L$$

| | |
|---|---|
| Sp. latent heat of ice | $3.36 \times 10^5$ J kg$^{-1}$ |

- **Specific latent heat** = heat to change 1 kg of a substance from solid to liquid at constant temperature.
- **Heating curve:** temperature rises, stays flat during melting (heat goes into breaking bonds), rises again.
- **Total heat for ice at −10 °C to water at 20 °C:** m $c_{ice}$ (10) + m L + m $c_{water}$ (20).

### Radioactivity

- Spontaneous disintegration of an unstable nucleus. Not affected by temperature or pressure.

| | α | β | γ |
|---|---|---|---|
| **Nature** | $^4_2$He nucleus | Electron | EM wave |
| **Charge** | +2 | −1 | 0 |
| **Penetration** | Lowest (paper) | Medium (Al sheet) | Highest (thick lead) |
| **Ionisation** | Highest | Medium | Lowest |
| **Deflected by field** | Yes (to −) | Yes (to +) | No |

| | |
|---|---|
| Atomic number | Z = protons |
| Mass number | A = protons + neutrons |

### Before you write the answer

- **1.** Convert to SI units first (g → kg, cm → m, minutes → s, °C → K where needed).
- **2.** Write **Given**, **To find**, **Formula**, **Substitution**, **Answer + unit**.
- **3.** Draw a neat, **labelled** diagram (ray diagrams: arrows on rays, ruler lines, dashed for virtual).
- **4.** Use the **sign convention** for mirrors and lenses every time.
- **5.** Underline the final answer.

### Nuclear equations

$$\alpha \text{ decay: } {}^A_ZX \rightarrow {}^{A-4}_{Z-2}Y + {}^4_2He$$

$$\beta \text{ decay: } {}^A_ZX \rightarrow {}^A_{Z+1}Y + {}^0_{-1}e$$

Example (α): $^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^4_2He$

Example (β): $^{14}_6C \rightarrow {}^{14}_7N + {}^0_{-1}e$

- **Uses:** carbon dating, cancer treatment, sterilisation, tracers. **Harm:** burns, cancer, genetic damage.
- **Safety:** lead shielding, gloves, tongs, short exposure, distance. **Background radiation:** cosmic rays, rocks, radon, X-rays.

### Fission & fusion

| | Fission | Fusion |
|---|---|---|
| **Process** | Heavy nucleus splits | Light nuclei join |
| **Example** | U-235 | Hydrogen isotopes |
| **Conditions** | Slow neutron | Very high T and P |
| **Used in** | Reactors, atom bomb | Sun, H-bomb |

$$^{235}_{92}U + {}^1_0n \rightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3\,{}^1_0n + \text{energy}$$

$$^2_1H + {}^3_1H \rightarrow {}^4_2He + {}^1_0n + \text{energy}$$

## 11 Score-saver checklist

### 30-second revision map

| Chapter | The one formula |
|---|---|
| **Moments** | F × d |
| **Work** | F s cos θ |
| **Machine** | η = MA/VR × 100 |
| **Refraction** | μ = sin i / sin r |
| **Lens** | 1/v − 1/u = 1/f |
| **Sound** | v = fλ; 2d = vt |
| **Ohm** | V = IR |
| **Power** | P = VI |
| **Transformer** | $V_s/V_p = N_s/N_p$ |
| **Heat** | Q = mcΔT; Q = mL |

Saviours AI

# That was the cheat sheet. Now go practise.

Formulas only help when you can use them under exam pressure. Saviours AI is built only for ICSE Class 10: solve doubts, plan your prep, test yourself and stay focused.

### Solve doubts

Stuck on a numerical or an equation? Get it explained step by step.

### Test yourself

Turn what you just revised into questions and find your weak spots.

### Plan & focus

A prep plan for your boards and a distraction-free study loop.

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